

Advait Parab

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Education

Sardar Patel Institute of Technology, Mumbai

2023 – 2027

– B.Tech in Computer Engineering

CGPA: 9.03/10

Experience

Mahindra Finance — AI Intern (DnA)

Summer 2026

– Building an enterprise **agentic AI voice coaching assistant** that automates performance coaching, conversational intelligence, actionable insight generation, and managerial reporting for sales teams.

Projects

TerrainAI — Autonomous Rover Terrain Perception 🌐

Mar 2026

- Developed a **real-time semantic segmentation pipeline** for autonomous rover navigation using **FPN+MiT-B3, DeepLabV3++EfficientNet-B4, and LinkNet+MobileNetV2**, with dynamic model switching driven by **Lucas-Kanade optical flow**.
- Designed a **14-stage augmentation pipeline**, composite **Focal + Dice + Boundary Loss, GradCAM/SHAP explainability**, and **A* risk-aware path planning**; deployed via **FastAPI** with live inference visualization.

Backdoor Attack Detection in Large Language Models

Present

- Studied semantic backdoors in **QLoRA-fine-tuned LLMs** by injecting domain-triggered sentiment and jailbreak behaviors into **Qwen2.5-0.5B-Instruct**, demonstrating stealthy manipulation without degrading general performance.
- Proposed grey-box and strictly black-box detection methods based on **cross-perplexity separation, excess divergence, and contrastive sentiment shift**, achieving **100% detection accuracy** while reducing audit scope from **15 domains to 2–3 candidate triggers**.

F1 Analytics & Prediction Platform 🌐

2025

- Designed a full-stack Formula 1 analytics platform featuring historical race intelligence, custom REST APIs, database-backed data pipelines, and interactive performance dashboards.
- Built machine learning pipelines using **CatBoost, XGBoost, and Random Forest** models to predict qualifying positions and race outcomes, incorporating driver, constructor, circuit, and season-level features.

Atria — Multi-Agent AI System (EY Hackathon) 🌐

2025

- Built a **12+ agent AI framework** that combined clinical, patent, research, and market intelligence sources to identify drug repurposing candidates through collaborative agent reasoning.
- Implemented confidence scoring, hallucination control, retrieval pipelines, and model fallback strategies to improve output reliability, with results visualized through a **Next.js dashboard**.

Positions of Responsibility

IEEE Computer Society, SPIT — Vice Chairperson

2024 – 2025

- Managed event operations, sponsorship coordination, and backend systems for a **24-hour AI/ML hackathon**, supporting participant onboarding and event execution.

Oculus Cube Open & Mumbai Cubing Sprint — Lead Organizer

2024 – Present

- Led finances, logistics, and execution for **3 official Rubik's Cube (WCA) competitions**, coordinating venues, registrations, and competition operations.

Technical Skills

- **Languages:** Python, Java, C++, JavaScript, SQL
- **AI/ML:** Scikit-learn, TensorFlow, PyTorch, OpenCV, NLP, Computer Vision
- **Frameworks & Backend:** FastAPI, Flask, Node.js, Express.js, PostgreSQL, MySQL, MongoDB
- **Tools & Technologies:** Git, Docker (basic), Postman, VS Code

Achievements

- **TenzorX 2026 National AI Hackathon (Poonawalla Fincorp) — Top 10 Finalist:** Top 10 among **20,000+ teams**.
- **CSI SPIT Hackathon — Domain Winner (AI/ML):** Best solution in the AI/ML track.
- **Aether HackConquest — 1st Runner-Up:** MFA-enabled password vault with voice, OTP, and QR authentication.
- **Commit 1.0 Hackathon — 2nd Runner-Up:** Gamified AES-256 password vault with zxcvbn validation.
- **Google Praxis 2.0 — Finalist:** Nationwide GenAI + ML hackathon finalist.
- **Pragati AI for Impact Hackathon (Meta x The/Nudge) — Finalist:** Scalable GenAI/ML solution for social impact.